

CIS241

System-Level Programming and Utilities

git (intro)

Erik Fredericks, frederer@gvsu.edu

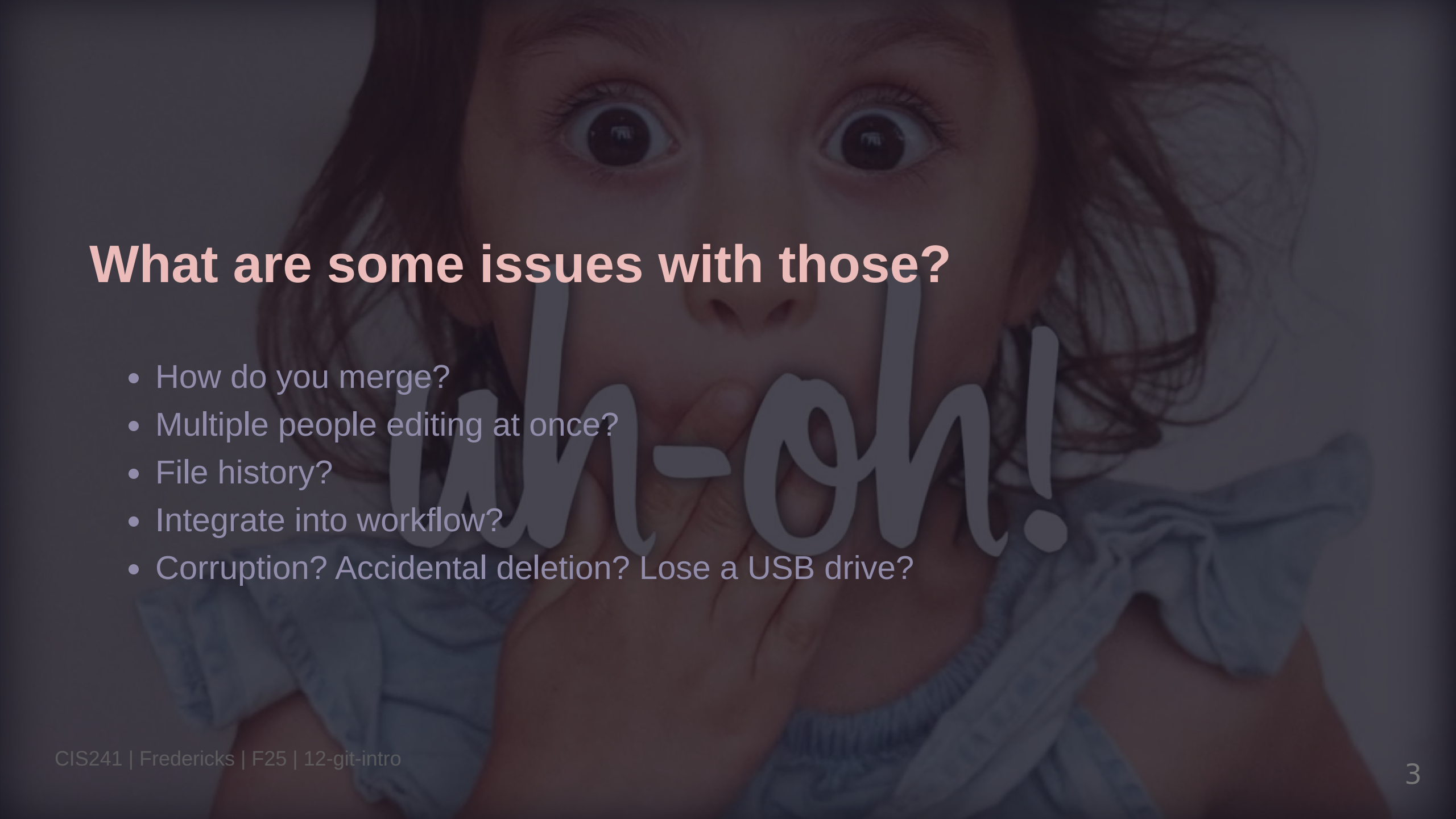
Fall 2025

Based on material provided by Erin Carrier, Austin Ferguson, and Katherine Bowers

How do you share files?

Or backup/transfer files?

Or manage a group project?



What are some issues with those?

- How do you merge?
- Multiple people editing at once?
- File history?
- Integrate into workflow?
- Corruption? Accidental deletion? Lose a USB drive?



Source control!

Common options:

- **git**
*
- subversion

Version control systems

Software that tracks all changes to a repository of files

Benefits:

- Full history
 - Rollbacks, traceability
- Working **concurrently**
- Merging
- Files never fully deleted

Distributed version control systems

- Each user has a local copy of *all* files
 - Including full history!
- Allows for working offline
- In theory, should make merging easier

Common version control systems

Distributed

- git
- mercurial

Centralized

- subversion
- perforce



Wot about GitHub?

git -- *local*

- software
- handles version control
- operates locally

GitHub -- *remote*

- Website/server
- Stores git repositories
- Additional features
 - Public sites
 - GitHub Actions

Touch of history on git

git:

- **The most widely used source code management tool: 42.9% of professional software developers use it as their primary source control system.**
 - 90% respondents say they use Git (2021 StackOverflow survey)
- **Developed by Linus Torvalds in 2005 for use developing the Linux kernel**
- **GitHub was developed in 2008, and provides free (and paid) Git repository hosting**
 - Currently owned by none other than Microsoft!

Personal notes

git is *complicated*

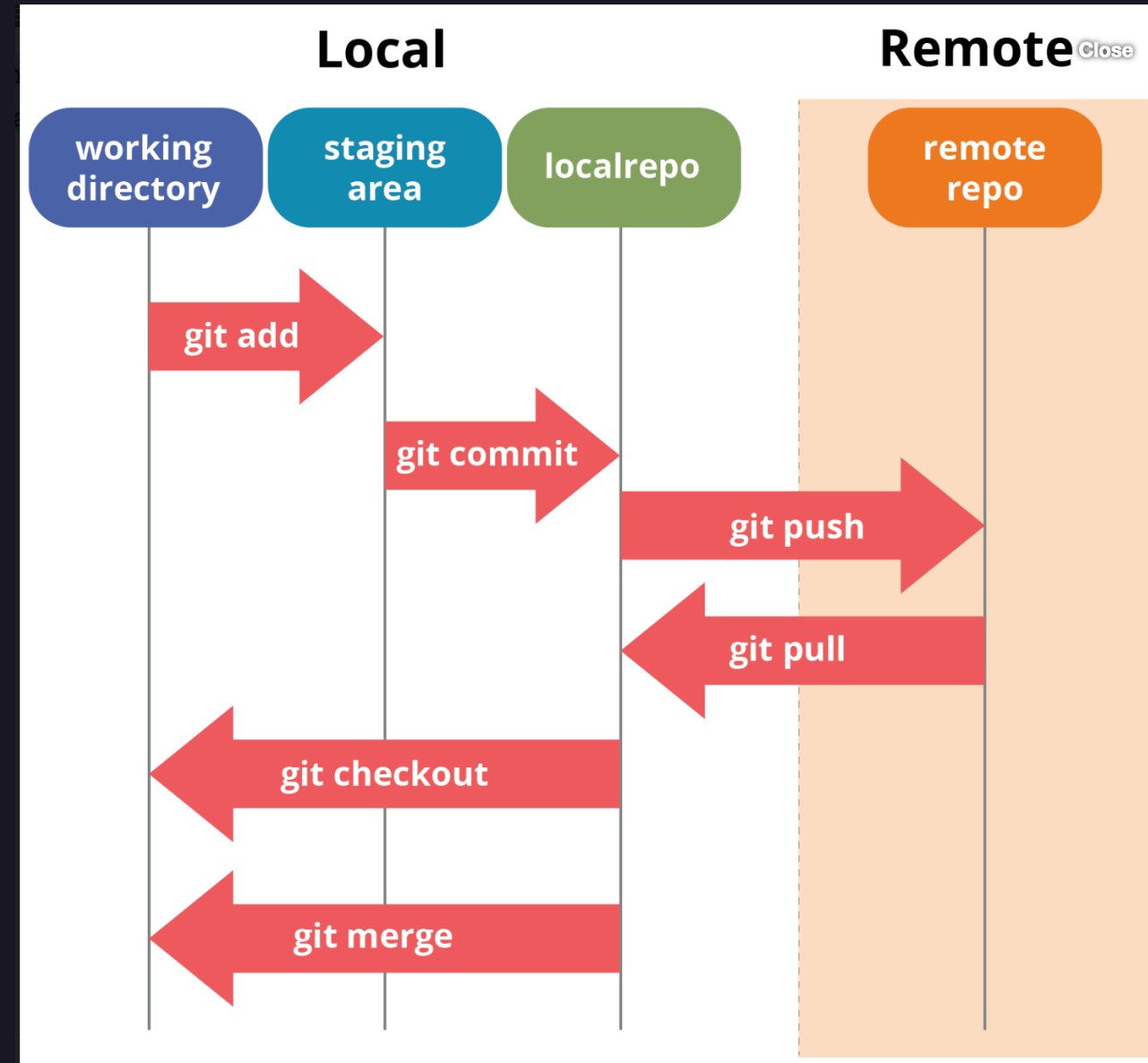
You will mess things up / break things

- I still do
- Most mistakes are recoverable
- Learn **how** to recover things!

Don't worry too much about the details

- Get comfortable with the big picture
- Look things up when you need (or are curious)

The process



git: repositories

Git repositories contain:

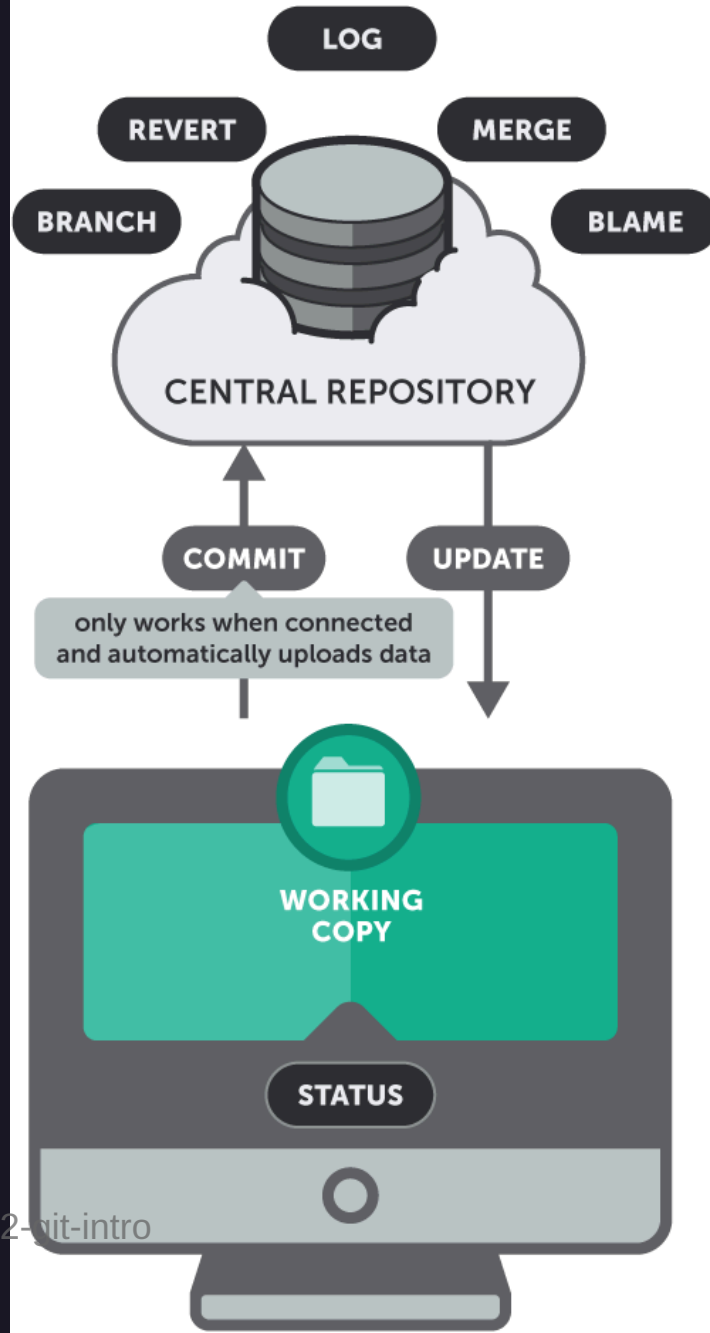
- A set of commit objects, and
- A set of references to commit objects (heads)

Git repositories are stored:

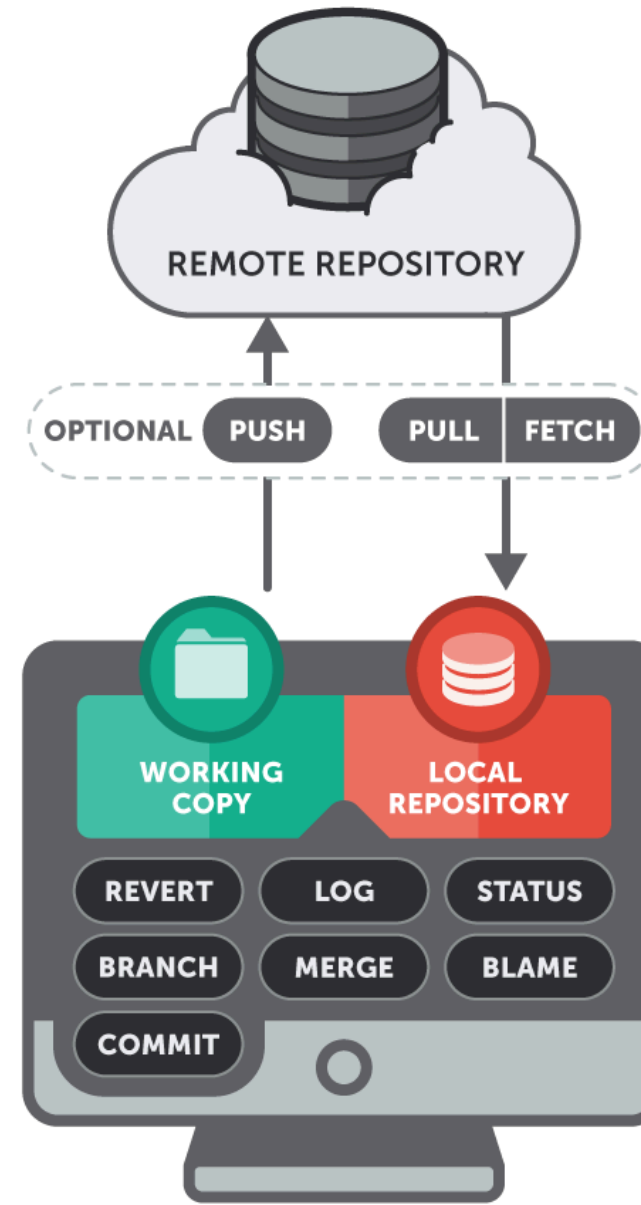
- In a .git sub-directory in the same directory as the project
- There is no central repository server

Git is distributed!

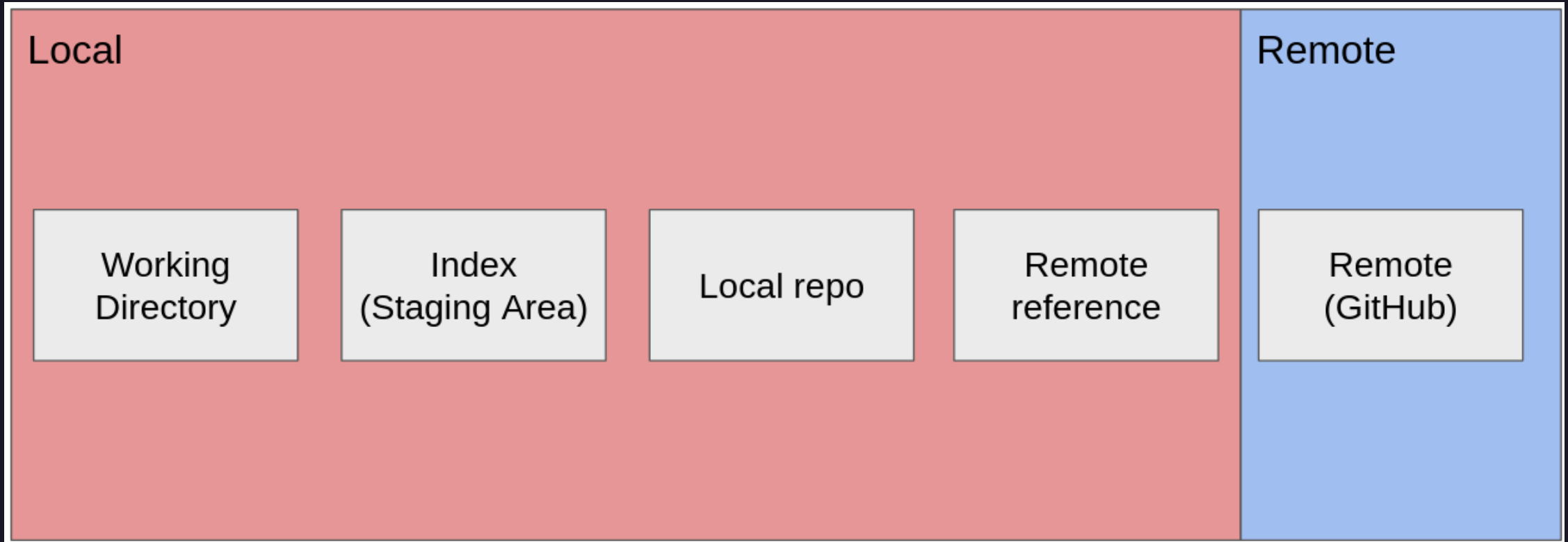
SUBVERSION



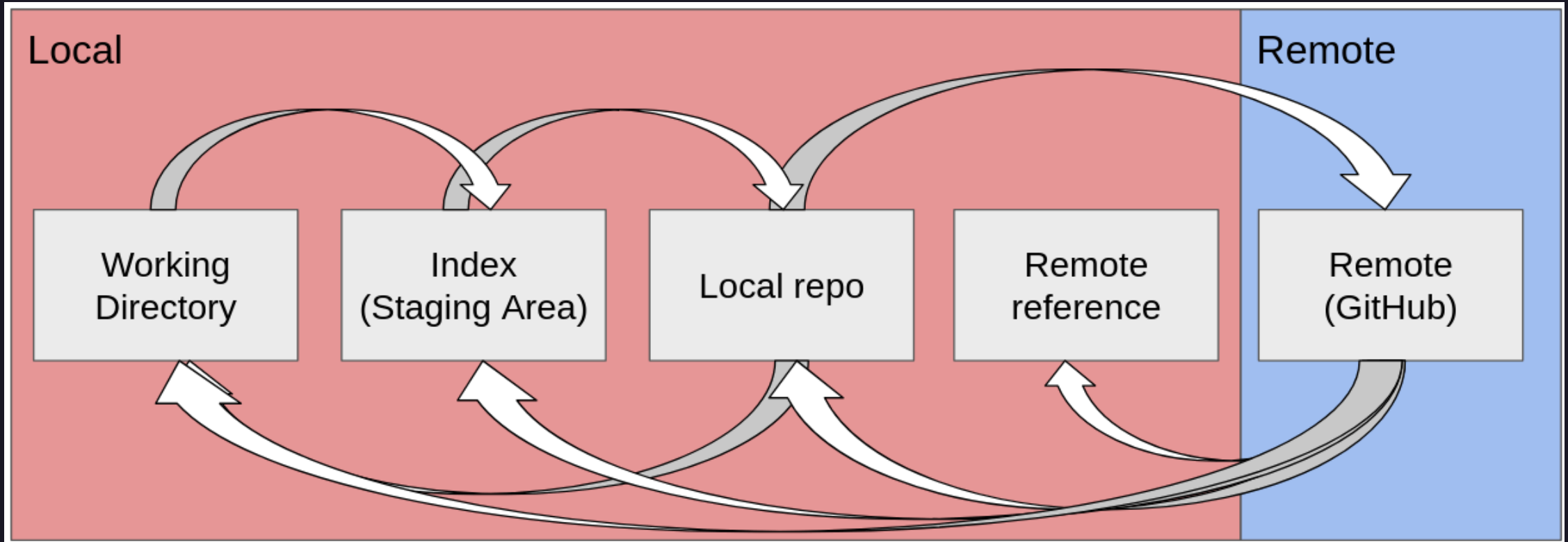
GIT



Where we're going



Where we're going



Today's work

1. Install git

- Probably already installed, but you never know
 - <https://github.com/git-guides/install-git>
 - We are **not** using GitHub Desktop! Terminal only!
 - To test, try `git version`

2. Create a GitHub account - you'll be sending me your username

3. Configure git on your machine

- `git config --global user.name "Your name"`
- `git config --global user.email "Your email"`